Lesson 4: Direct Hit

Scratch

VIDEO GAME CREATION

This week: detecting when things touch. Your job is to add one enemy to the screen and make your bullet destroy it on contact - the first piece of your capstone's actual combat loop.

Open your own [YourName]Game project from Lesson 3 - don't start a new one. Everything this week builds on the player and bullet you already have.

Tip: this lesson's Assets folder has a ready-to-use placeholder enemy image (enemy_invader_red.png) and a new sound effect (sfx_hit.wav) - you don't need to find or make your own art today.

Setup

1. Open your [YourName]Game project from Lesson 3.
2. Create a new sprite for the enemy - import enemy_invader_red.png from this lesson's Assets folder (Upload Sprite).
3. Rename this sprite Enemy. Add a "when green flag clicked" script to it with "go to x: 0 y: 150" (Motion) and "show" (Looks), so it starts in a fixed spot near the top of the stage.

Level 1 - Minimal (MVP)

A single enemy sits on screen. When your bullet touches it, both the bullet and the enemy disappear. This is a complete finish for today - reaching this is enough to move on to next week.

4. Switch to the Bullet sprite's "when I start as a clone" script (from Lesson 3).
5. Inside the same forever loop that moves the bullet upward, add "if <touching [Enemy]?> then" (Control + Sensing), and put "broadcast [enemy hit]" and "delete this clone" inside it - check touching before the existing off-screen check, so a hit near the top of the stage is still caught.
6. Click the New Message option in the broadcast dropdown to create the "enemy hit" message the first time.
7. Switch to the Enemy sprite, click Events, and drag "when I receive [enemy hit]" into a new script, with "hide" underneath it.
8. Click the stage, then press spacebar aimed at the Enemy to test - the enemy should vanish and the bullet that hit it should disappear too.

Level 2 - Complete (Alpha/Beta)

The enemy is destroyed cleanly and exactly once, even if you fire several bullets in a row - no errors, no lingering references, and a bullet that misses still behaves exactly like it did last week.

- Fire several bullets in a row, including some that miss - each hit should make the enemy vanish exactly once, and every bullet that misses should keep flying and disappear off-screen exactly like it did in Lesson 3.
- This already works with no extra code: Scratch's "touching" check always returns false for a hidden sprite, so once the Enemy sprite is hidden, no later bullet clone can trigger a second "enemy hit" broadcast, even with several bullets in flight at once.

Level 3 - Refined (Ready to Publish) - pick any that interest you

- Sound: upload sfx_hit.wav from this lesson's Assets folder, then add "start sound [sfx_hit]" to the Enemy's "when I receive [enemy hit]" script, right before "hide".
- Visual: give the Enemy its own distinct color using the Costume editor's Fill tool, or add a "set color effect to" block (Looks) in its green-flag script, so it clearly reads as a threat and not just another shape on screen.
- Structure: the Bullet's forever loop should only ever detect a hit and report it (touching check + broadcast + delete this clone) - everything about how the Enemy responds to being hit (sound, hiding) belongs on the Enemy's own "when I receive [enemy hit]" script. Keeping this boundary clean is what let the Sound idea above drop into the Enemy sprite alone.

If Something's Wrong

- Bullet passes through the Enemy without anything happening: confirm "touching [Enemy]?" actually targets the Enemy sprite by name in the dropdown (it can default to something else), and that the check is inside the bullet's forever loop, not outside it.
- Enemy disappears but a leftover bullet keeps flying: confirm "delete this clone" runs in the same if-branch as the touching check, right alongside the broadcast.
- Nothing happens even up close: confirm the Enemy sprite's own "when green flag clicked" script actually runs "show" - a sprite hidden from a previous test stays hidden until the green flag is clicked again.

Want More?

Student Resource List > Scratch section, or scratch.mit.edu/ideas